

Stock-Level GARCH + LSTM: Retraining and Economic Value

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Completed run; forecast robustness failed

Processed **503 stocks** and saved **1,968 LSTM fits** for forecasts dated 2015-07-01 through 2024-12-31. GARCH was also refitted from the raw return history; the earlier cached hybrid forecasts were not reused.

This is the separate stock-level multiplicative volatility model, not the index additive variance-residual LSTM in Part A. The studies are not one shared model.

What was trained

Daily GARCH(1,1)-t on 50 prior returns. An 8-unit LSTM predicts the ratio of trailing 21-day realized volatility to GARCH sigma using 10 past ratios. Expanding-history fitting uses 10 epochs, batch 8, and refits every 600 eligible forecast dates. Scaling and training labels stop strictly before each fit block.

Forecast quality on common observations

Forecast / trailing 21-day target	Date-equal MSE	MSE / GARCH
GARCH with recorded fit fallback	1.2789e-04	1.000x
GARCH x LSTM ratio (raw)	1.9130e-01	1495.801x
Previous trailing volatility	3.0968e-06	0.024x

1,165,385 common stock/date observations across 2,392 dates and 502 stocks. Daily losses average stocks first, then dates equally; lower MSE is better.

Hybrid MSE change versus GARCH: **+149480.14%** (worse). Hybrid-minus-GARCH loss interval: [1.020e-01, 3.052e-01]. The conditional interval is above zero.

Failure concentration: 493/502 stocks improve individually, yet AMCR contributes 99.96% of hybrid MSE. Near-zero GARCH denominators create extreme ratio targets. Independent checkpoint replay reproduces the failure; no stocks were removed.

Target caveat: consecutive 21-day volatility windows overlap. The persistence control tests this mechanical predictability; lower trailing-volatility error is not automatically better fresh future-risk prediction or trading performance.

Audit trail and limitations

Saved checkpoints, scalars, seeds, fit cutoffs and source hashes. 475 nonpositive raw hybrid outputs; 1 stock(s) have insufficient history for a neural fit. GARCH uses recorded past-volatility fallback on 63 evaluation asset-days.

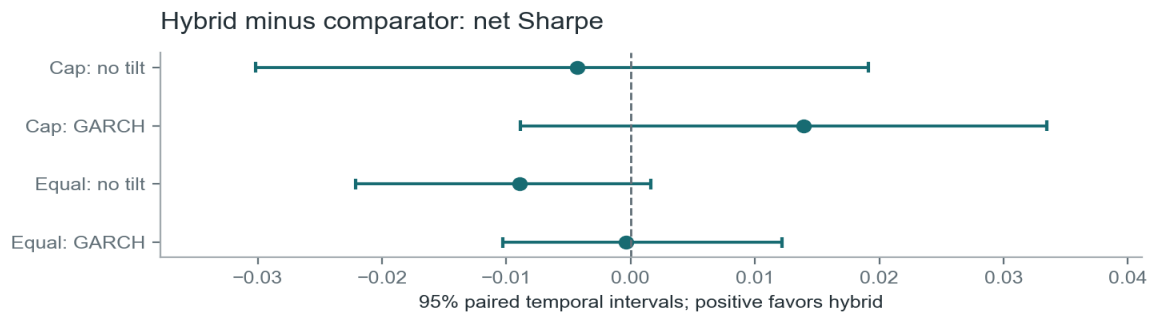
Later constituents/cap weights remain biased; one seed per fit and no untouched holdout. Reproducible model provenance does not establish robust forecasts or alpha.

Does the LSTM Add Trading Value?

Weight = base weight x clip(trailing volatility / forecast, 0.5, 1.5), normalized. Compare the hybrid with both an untilted allocation and a GARCH-only tilt.

Monthly, 10 bps	Net Sharpe	Max drawdown
Snapshot-cap base	1.119	-33.25%
Snapshot-cap + GARCH	1.101	-33.15%
Snapshot-cap + GARCH/LSTM	1.115	-32.48%
Equal-weight base	0.738	-38.34%
Equal-weight + GARCH	0.729	-37.79%
Equal-weight + GARCH/LSTM	0.729	-38.15%

2,390 common return dates, 2015-07-06 to 2024-12-31. Next-close execution; 10 bps per dollar bought/sold; drifting holdings and self-financing costs. Sharpe assumes daily risk-free return 0.0001.



Snapshot-cap hybrid minus matched GARCH Sharpe: +0.0139; 95% interval [-0.0089, +0.0335]. The conditional interval includes zero. Mean bootstrap block 21 days, 2,000 paired resamples.

Matched controls and implementation

Both tilts use the same stock pool and model-availability dates. Unavailable or nonpositive forecasts receive neutral tilts. Unrestricted GARCH and all predefined frequency/cost sensitivities remain in the aggregate results.

What this does not establish

Forecast robustness failed: 475 nonpositive outputs and AMCR-driven extreme errors. Capping portfolio tilts does not repair the underlying forecast model.

Later constituents and repeated later market caps create survivorship and look-ahead bias. Equal weights do not fix the universe. These conditional comparisons are not validated alpha, achievable-return estimates or a causal proof of LSTM value.

Full report (8 pages) | Code, all sensitivities and source hashes

Source: reports/stock_retraining_results.json. Raw vendor data and per-stock model artifacts remain local. Based on collaborative research; report prepared by Eddie Liu.